

Oleksandr Tomalak, Ph.D.

Theoretical and Computational Physics

Nuclear and Particle Physics

6312, Institute of Theoretical Physics
Chinese Academy of Sciences
No. 55 Zhong Guan Cun East Road
Haidian, Beijing, P. R. China
☎ (+86) 157 01075139
☎ (+49) 155 60735486
✉ tomalak@itp.ac.cn
📁 tomalak7.github.io
🌐 Oleksandr Tomalak

Experience

- since 2025 **Associate Professor**, Institute of Theoretical Physics, Chinese Academy of Sciences, Beijing, P. R. China.
- **Advanced Research in Neutrino, Hadron, and Nuclear Physics** resulting in collaborative grants, mentoring and communication skills as
- 2021–2024 **Director's Postdoctoral Fellow**, Los Alamos National Laboratory, NM, USA.
- **Bringing Radiative Corrections and Modern Nucleon Structure to Neutrino Physics**, mentoring and communication skills as
- 2018–2021 **Postdoctoral researcher**, University of Kentucky, Lexington, KY, USA and Fermilab, Batavia, IL, USA.
- **Theory of Atomic Physics and Lepton Scattering**, mentoring skills as
- 2016–2018 **Postdoctoral researcher**, Johannes Gutenberg-Universität Mainz, Mainz, Germany.
- **Precise Nucleon Structure**, teaching and communication skills as
- 2012–2016 **Ph.D. student**, Johannes Gutenberg-Universität Mainz, Mainz, Germany.
- **Analytical and Numerical Calculations in Early Universe Cosmology** as
- 2012–2015 **Ph.D. student**, Taras Shevchenko National University, Kyiv, Ukraine.
- **Programming and advanced communication skills** as
- 2015–2017 **Guest scientist**, Department of Physics, University of Pavia, Pavia, Italy.
- **Data analysis, detector monitoring, and data taking** as
- 2010–2011 **Member of D0 Collaboration**, D0 Experiment, Fermilab, Batavia, IL, USA.
- **Calibration of detectors in high-energy physics experiment** as
- 2009–2010 **Bachelor student**, ZEUS Experiment, DESY, Hamburg, Germany.

Computer Skills

- | | | | |
|-----------------------|-------------------|-------------------------------|------------|
| - Wolfram Mathematica | - LaTeX | - C, C++ | - Bash |
| - Maple | - Origin | - Fortran | - Batch |
| - Mathcad | - DataGraph | - Python, R | - Root |
| - Windows | - Linux | - OS X | - JaxoDraw |
| - HTML, CSS | - SQL | - JavaScript | - Jinja |
| - Flask, Django | - office software | - video conferencing software | |

Languages

English	Fluent	C1 level (JGU Mainz, 2018)
German	Upper intermediate	C1 level (JGU Mainz, 2018)
Russian	Fluent	
Ukrainian	Fluent	
Mandarin	Elementary	
Chinese		

Education

- 2012–2016 **Ph.D. in Theoretical Physics**, Johannes Gutenberg-Universität Mainz, Mainz, Germany, *Summa Cum Laude*.
- 2010–2012 **MS in Theoretical Physics**, Taras Shevchenko National University, Kyiv, Ukraine, *Red Diploma*.
Speciality: Nuclear and Particle Physics, Quantum Field Theory
- 2006–2010 **BS in Physics**, Taras Shevchenko National University, Kyiv, Ukraine, *Red Diploma*.
Speciality: Nuclear and Particle Physics, Quantum Field Theory

Professional Activities

Organized 8 Meetings and Conferences
Refereed for 16 Journals
Mentored 7 Students and 2 Postdocs
Guest editor for *Universe*
Wrote 10 Collaborative Research Proposals: 6 Proposals funded
Support from 6 Research Fellowships

73 Peer-Reviewed Journal Articles

The up-to-date publication and citation information can be found on the web page:
<https://inspirehep.net/authors/1077543>
ORCID: 0000-0002-4827-5842

180 Professional Presentations

58 Invited Talks
111 Contributed Talks
11 Posters at International Conferences

Selected Awards and Honors

- 2020–2021 Universities Research Association (URA) Salary Support, Fermilab, USA
- 2019 Universities Research Association (URA) Visiting Scholars Program, Fermilab, USA
- 2017 Prize for Outstanding Doctoral Research, Faculty 08: Physics, Mathematics and Computer Science, Johannes Gutenberg-Universität Mainz, Mainz, Germany
- 2012 Alumnus of the Year, Taras Shevchenko National University, Kyiv, Ukraine
- 2008 3rd degree diploma, member of the Faculty Team, Ukrainian Physicists' Tournament for University Students, Kyiv, Ukraine
- 2007 2nd degree diploma, member of the Faculty Team, Ukrainian Physicists' Tournament for University Students, Kyiv, Ukraine
- 2007 Absolute Winner of Ukrainian Students Olympiad in Theoretical Physics, Kharkiv, Ukraine
- 2006 3rd degree diploma, member of the Faculty Team, Ukrainian Physicists' Tournament for University Students, Kyiv, Ukraine
- 2006 Silver Medal, 37th International Physics Olympiad, Singapore



May 14, 2026